Teleportation Protocol

1. Alice sends her qubit through CNOT gate
2. Alice then sends the first qubit through a Hadamard gate
3. Alice measures her qubits, obtaining one of four results and send this information to Bob.
4. Given Alice's measurements Bob performs one of four operations on his half of EPR pair and recovers original quantum state

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

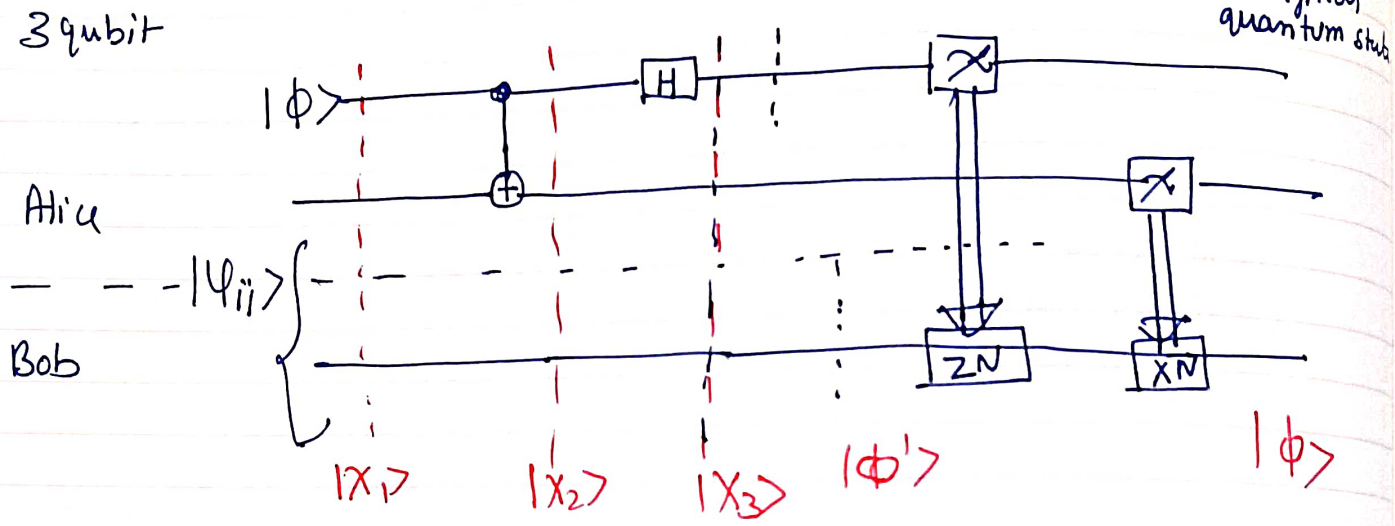
$- - - |\varphi_{ij}\rangle -$

first part of bell state (one qubit)

second part of bell state (one qubit)

$|\varphi_{ij}\rangle$

3 qubit



$$|\varphi_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|X_1\rangle = |\phi\rangle \otimes |\varphi_{00}\rangle$$

$$= (\alpha|0\rangle + \beta|1\rangle)\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$= \frac{1}{\sqrt{2}}(\alpha|0\rangle)(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)$$

Now apply CNOT

first qubit, second qubit

$$|X_2\rangle = \frac{1}{\sqrt{2}}(\alpha|0\rangle|00\rangle + |11\rangle) + \beta|1\rangle(|10\rangle + |01\rangle)$$

Hadamard gate on first qubit

$$|X_3\rangle = \frac{1}{\sqrt{2}}\left(\alpha\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)(|10\rangle + |01\rangle)\right)$$

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$$= \frac{1}{2}\Big(\alpha(|000\rangle + |011\rangle + |100\rangle + |111\rangle) + \beta(|010\rangle + |001\rangle - |110\rangle - |101\rangle)\Big)$$

$$= \frac{1}{2}\Big(|00\rangle\,(\alpha|0\rangle + \beta|1\rangle) + |01\rangle\,(\alpha|1\rangle + \beta|0\rangle) + |10\rangle\,(\alpha|0\rangle - \beta|1\rangle) + |11\rangle\,(\alpha|1\rangle - \beta|0\rangle)\Big)$$

$|11\rangle \downarrow$ Alice (first two qubit)

$(\alpha|1\rangle - \beta|0\rangle) \downarrow$ Bob

Alice's Measurement MN	Bob's state $\lvert\phi'\rangle$
00	$\alpha\lvert 0\rangle + \beta\lvert 1\rangle$
01	$\alpha\lvert 1\rangle + \beta\lvert 0\rangle$
10	$\alpha\lvert 0\rangle - \beta\lvert 1\rangle$
11	$\alpha\lvert 1\rangle - \beta\lvert 0\rangle$

<u>How to get $|\phi\rangle$ from $|\phi'\rangle$ using MN :</u>

MN = 00 ⇒ Do nothing	$-\boxed{Z^0}-\boxed{X^0}-$	$\alpha\lvert 0\rangle + \beta\lvert 1\rangle$
MN = 01 ⇒ Bit flip (X-gate)	$-\boxed{Z^0}-\boxed{X^1}-$	$\alpha\lvert 1\rangle + \beta\lvert 0\rangle$
MN = 10 ⇒ Phase flip (Z-gate)	$-\boxed{Z^1}-\boxed{X^0}-$	$\alpha\lvert 0\rangle - \beta\lvert 1\rangle$
MN = 11 ⇒ Bit and Phase flip (X and Z-gate)	$-\boxed{Z^1}-\boxed{X^1}-$	$\alpha\lvert 1\rangle - \beta\lvert 0\rangle$

The teleportation protocol can also be completed using $|\varphi_{01}\rangle$, $|\varphi_{10}\rangle$ or $|\varphi_{11}\rangle$.

$\downarrow$

Teleportation has been experimentally verified.

X. Rotation Gates:-

The Rotation gates are defined as follows:

$$R_x(\theta) \equiv e^{\frac{-i\theta X}{2}} \; ; \; R_y(\theta) \equiv e^{\frac{-i\theta Y}{2}} \; ; \; R_z(\theta) \equiv e^{\frac{-i\theta Z}{2}}$$

we know $X^2 = I$, $Y^2 = I$ and $Z^2 = I$,

we can write the rotation gates as:

$$R_x(\theta) \equiv e^{\frac{-i\theta X}{2}} = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) X$$

$$R_y(\theta) \equiv e^{\frac{-i\theta Y}{2}} = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) Y$$

$$R_z(\theta) \equiv e^{\frac{-i\theta Z}{2}} = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) Z$$

We know

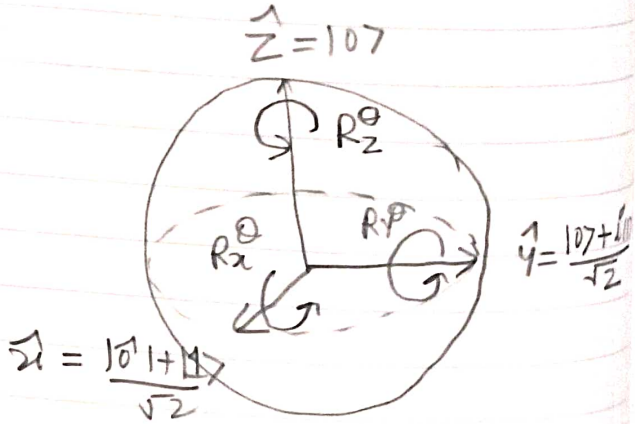
$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \; ; \; X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \; ; \; Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \; ; \; Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

We can now write the rotation gates as matrices in the computational basis:

$$R_x(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\theta/2\right) \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\frac{\theta}{2} \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} e^{\frac{-i\theta}{2}} & 0 \\ 0 & e^{\frac{i\theta}{2}} \end{bmatrix}$$



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$$|q_1\rangle \longrightarrow \boxed{R_x^{\theta}} \longrightarrow |q_2\rangle = R_x^{\theta}|q_1\rangle$$

or $R_x(\theta)\,|q_1\rangle$

$$|q_1\rangle \longrightarrow \boxed{R_y^{\theta}} \longrightarrow |q_2\rangle = R_y^{\theta}|q_1\rangle$$

$$|q_1\rangle \longrightarrow \boxed{R_z^{\theta}} \longrightarrow |q_2\rangle = R_z^{\theta}|q_1\rangle$$